```python
import numpy as np

#sum of element of 3 x 3
arr3d = np.random.randint(1,11,size=(3,3))
print(arr3d)
Sum = np.sum(arr3d)
print("Sum of all Elements : ",Sum)
```

```
[[ 2  5  2]
 [ 4  9  4]
 [ 2  6 10]]
Sum of all Elements :  44
```

```python
#avg of element of 4 x 4
arr4d = np.random.rand(4,4)
print(arr4d)
avg = np.mean(arr4d)
print("Average of all Elements : ",avg)
```

```
[[0.31008133 0.30666187 0.03657576 0.03351542]
 [0.40332796 0.04365721 0.43461072 0.13389531]
 [0.86378715 0.21645602 0.27535305 0.07617275]
 [0.8475716  0.15397093 0.68304676 0.56588243]]
Average of all Elements :  0.33653539215167105
```

```python
#sum of element of 2 x 3
arr2x3 = np.random.rand(2,3)
print(arr2x3)
Sum = np.sum(arr2x3[:,1])
print("Sum of all Elements of 2nd row : ",Sum)
```

```
[[0.62572517 0.34119676 0.20156804]
 [0.80421055 0.28441033 0.87016062]]
Sum of all Elements of 2nd row :  0.6256070889323414
```

```python
# find highest element in 5x5
arr5x5 = np.random.randint(1,10001,size=(5,5))
print(arr5x5)
Max = np.max(arr5x5)
print("Max among all Elements : ",Max)
```

```
[[1508 4004 1511 6881 7283]
 [ 530 7680  213 7470 5627]
 [1296 2579 2639 3159 1268]
 [1358 4418 7124  343 2146]
 [2876 8985  344  278 2659]]
Max among all Elements :  8985
```

```python
# find trace in 4x4
arr4x4 = np.random.rand(4,4)
print(arr4x4)
```

```
tr = np.trace(arr4x4)
print("sum of all diagonal elements : ",tr)
```

```
[[0.38969828 0.32538589 0.29082692 0.38099127]
 [0.87866076 0.99828621 0.93642776 0.59968742]
 [0.3412576  0.03916221 0.23970246 0.04576408]
 [0.48057372 0.27434913 0.94587712 0.26478667]]
sum of all diagonal elements :  1.8924736319021438
```

```
# find highest element in 3x3
arr3x3 = np.random.rand(3,3)
print(arr3x3)
Min = np.min(arr3x3)
print("Minimum among all Elements : ",Min)
```

```
[[0.38374901 0.23625982 0.61691999]
 [0.14352691 0.03455422 0.05900485]
 [0.64946168 0.99616589 0.63250666]]
Minimum among all Elements :  0.03455422138235509
```

```
# sum all elements > 10 in 3X3
arr3x3 = np.random.randint(1,21,size=(3,3))
print(arr3x3)
Sum = np.sum(arr3x3[arr3x3 > 10])
print("Sum of all Elements of 2nd row : ",Sum)
```

```
[[17  5  1]
 [13  1  4]
 [10  6 20]]
Sum of all Elements of 2nd row :  50
```

```
# Generate a 5x5 matrix of random numbers and find the mean of the
maximum values of each row.
arr5x5 = np.random.randint(1, 101, size=(5, 5))
print(arr5x5)
row_maxima = np.max(arr5x5, axis=1)
print("Row Max :", row_maxima)
mean_of_maxima = np.mean(row_maxima)
print("Mean of Max :", mean_of_maxima)
```

```
[[37 73 95 60 10]
 [21 89 84 70 62]
 [73 34 42 12 89]
 [24 71  6 20 25]
 [ 2 63 97 58 16]]
Row Maxima: [95 89 89 71 97]
Mean of Maxima: 88.2
```

```
# Create a 3x3 matrix and calculate the difference between the sum of
the elements in the first row and
# the sum of the elements in the last row.
```

```
arr3x3 = np.random.randint(1, 21, size=(3, 3))
print(arr3x3)
sum_first_row = np.sum(arr3x3[0, :])
sum_last_row = np.sum(arr3x3[-1, :])
difference = sum_first_row - sum_last_row
print("Difference between sum of first and last row : ", difference)
```

```
[[ 9 19 15]
 [18  5  5]
 [14  9 16]]
Difference between sum of first and last row :  4
```

```
# Generate a 6x6 matrix of random integers and replace all values
below the median with zero.
arr6x6 = np.random.randint(1, 101, size=(6, 6))
print(arr6x6)
median_val = np.median(arr6x6)
print("Median value : ",median_val)
arr6x6[arr6x6 < median_val] = 0
print("Matrix after replacing values below median with zero:\n",
arr6x6)
```

```
[[18 53  2 18 50 81]
 [12 16 15 92 29 77]
 [58 48 29 36 57 80]
 [33 69 16 18 48  9]
 [16 51 16 96 70 96]
 [56 25 44 28  7 47]]
Median value :  40.0
Matrix after replacing values below median with zero:
 [[ 0 53  0  0 50 81]
 [ 0  0  0 92  0 77]
 [58 48  0  0 57 80]
 [ 0 69  0  0 48  0]
 [ 0 51  0 96 70 96]
 [56  0 44  0  0 47]]
```

```
# Create a 4x4 matrix and compute the sum of elements in the upper
triangular part of the matrix.
arr4x4 = np.random.randint(1, 101, size=(4, 4))
print(arr4x4)
upper_tri_elements = arr4x4[np.triu_indices(4)]
sum_upper_tri = np.sum(upper_tri_elements)
print("Sum of elements in the upper triangular part : ",
sum_upper_tri)
```

```
[[90 84 62 47]
 [ 8 97 73 54]
 [97 85 61 23]
```

```
 [33 95 90 53]]
Sum of elements in the upper triangular part :  644
```

```
# Generate a 3x4 matrix of random values and find the standard
deviation of the elements in the third col.
arr3x4 = np.random.randint(1, 101, size=(3, 4))
print(arr3x4)
third_column_std = np.std(arr3x4[:, 2])
print("Standard deviation : ", third_column_std)
```

```
[[19 65 16 88]
 [12 74 19 34]
 [69 18 89 82]]
Standard deviation :  33.7276675083759
```